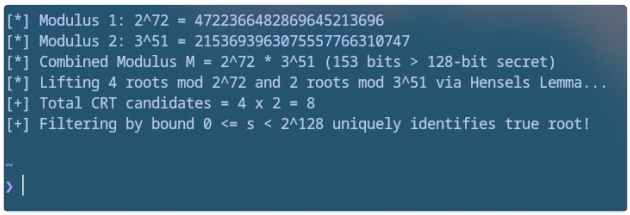
# Hensel Garden

## Cryptography




I inspected chall.py and data.json. A 128-bit secret seed s was shifted and squared under two coprime prime powers: $(s+87)^2 \bmod 2^{72}$ and $(s+143)^2 \bmod 3^{51}$.

The combined modulus $M = 2^{72} \times 3^{51} \approx 2^{152.8}$ bits strictly exceeded the 128-bit secret space, ensuring that the true seed could be uniquely identified via CRT.

I implemented 2-adic Hensel lifting modulo $2^{72}$ (yielding 4 candidate roots) and standard Hensel lifting modulo $3^{51}$ (yielding 2 roots). Combining all 8 pairs via the Chinese Remainder Theorem and filtering by $0 \le s < 2^{128}$ isolated the true root.

Finally, I tested the recovered flag against the executable / service:

**python3 Hensel_Garden.py**

Seed recovered: 282779946134366997745983790016833981964

**Flag Captured**

**SecLeaf{r00ts_gr0w_through_pr1me_p0w3rs}**

**BY:JOKER_53**